

A wolf is silhouetted against a misty, blue-toned forest background, howling with its head tilted back. The forest is dense with tall, thin trees and fog or mist hanging between them.

Facial rash and painful fingers

Systemic lupus erythematosus

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LKS Faculty of Medicine
23rd February 2024



lupus, lup- +

(Latin: wolf [pertaining to or connected with a "wolf"])

A fronte praecipitium a tergo lupi.

A precipice in front, wolves behind.

Equivalent to: "Between a rock and a hard place." "Between the devil and the deep blue sea."



Once upon a time there were 3 little pigs, so I ate them. End of story. Now go to sleep!

Lecture outline

- Case study – systemic autoimmunity
- What is SLE and why is it important?
- Classification criteria for SLE
 - Why are classification criteria useful / not useful?
- Clinical features of SLE
- The use of serological tests in SLE
- Treatment of SLE
- Prognosis of SLE

Core knowledge

There is much more information than needed for undergraduate students in this lecture handout. Students are expected to learn:

1. The immunopathogenesis of SLE
2. Evolution of the classification criteria of SLE based on
 - Systemic disease
 - Common disease manifestations
 - Demonstration of autoimmunity
3. Common clinical features of SLE
4. Interpretation of serological tests in SLE diagnosis and management
5. General management principles
6. General use of immunosuppressive drugs
7. Changing prognosis of SLE

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Helen F/26 years Secretary

- Presented 8 months ago with painful MCP and PIP arthritis of both hands associated with morning stiffness
- Diagnosed to have RA and treated with naproxen 250 mg t.d.s. with improvement
- Went to Bangkok 2 months ago and has been unwell since then

Helen F/26 years Secretary

- Admitted to Queen Mary Hospital
 - malaise
 - high fever
 - facial rash
 - lower limb purpuric petechiae
 - arthritis



Helen F/26 years Secretary

- Investigations

– WBC	$3.5 \times 10^9/l$ [4-11]
– Hgb	9.8 g/dl [11.5-16]
– Platelet	$54 \times 10^9/l$ [150-400]
– Reticulocyte	6%
– ESR	58 mm/hour

Helen F/26 years Secretary

- Investigations

– Urea	6.6 mmol/l [3.3-6.6]
– Creatinine	125 µmol/l [62-124]
– Bilirubin	45 mmol/l [0-17]
– Liver enzymes	normal
– Serum albumin	30 g/l [36-50]
– LDH	340 U/l [55-140]
– Urine P/Creat ratio	70 mg/mmol [<15]

Helen F/26 years Secretary

- Investigations

- ANA 1/1080
- Anti-ds DNA 230 U/L [<35]
- Anti-ENA Anti-Sm +ve, Anti-Ro +ve
- ACA (IgG) +ve
- Serum C3 0.56 g/l [0.9-1.8]
- Serum C4 0.12 g/l [0.1-0.5]
- Direct Coomb's test +ve
- Bone marrow exam Peripheral consumption

Dx: Systemic lupus erythematosus

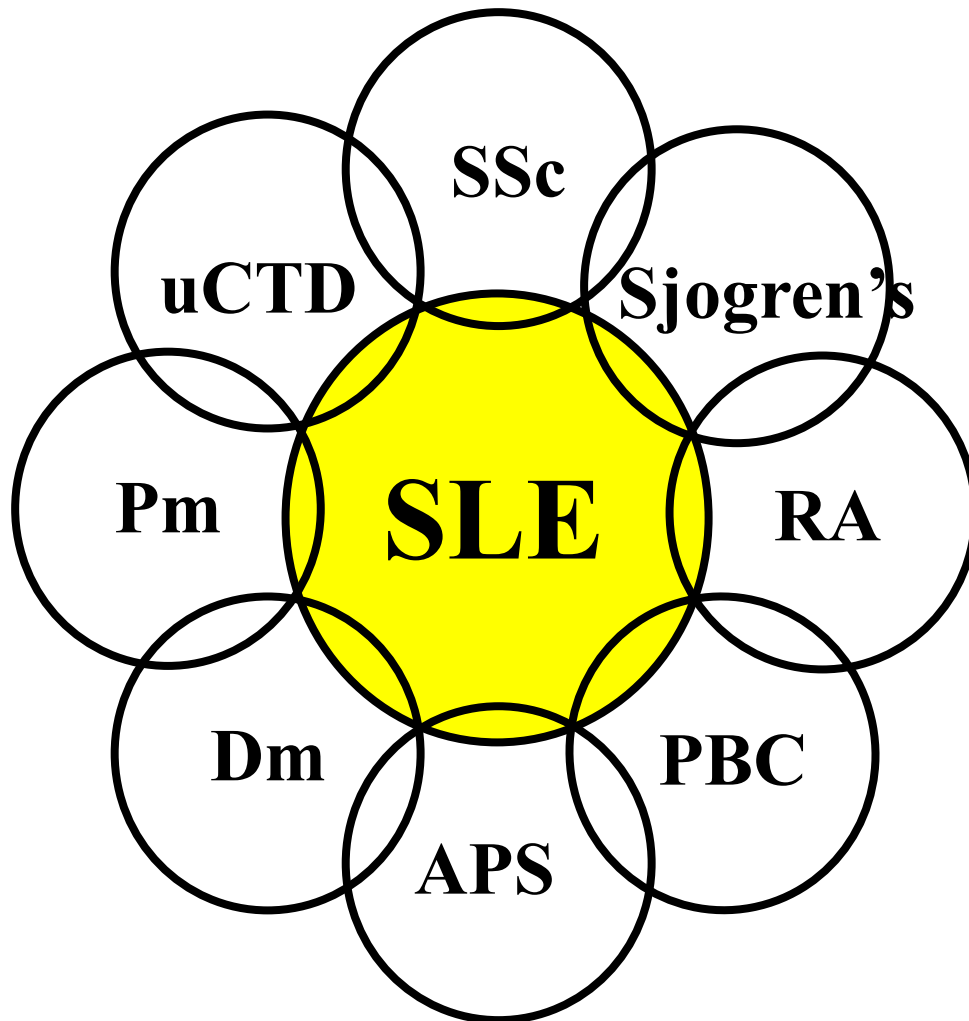
Salient features

- Young female
- Multi-system disease
 - Constitutional symptoms
 - Inflammatory polyarthrititis
 - Skin rashes
 - Haemic involvement
 - Renal involvement
- Possible triggering factor
 - Trip to Thailand
- Autoimmunity
 - Positive autoantibody screen
 - Coomb's test
- Immune complex disease
 - Lowered complement levels

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SLE – part of the spectrum of autoimmune rheumatic diseases



APS - Anti-phospholipid antibody syndrome

Dm - Dermatomyositis

PBC - Primary biliary cirrhosis

Pm - Polymyositis

RA - Rheumatoid arthritis

SSc - Systemic sclerosis

uCTD - Undifferentiated connective tissue disorder

SLE : Definition

- Systemic lupus erythematosus (SLE) is a chronic *multisystem autoimmune* disorder. It is characterised by profound immunological disturbances resulting in an excessive production of *autoantibodies*, some of which may cause direct tissue damage while others take part in *immune complex* (IC) formation. *Failure to remove these ICs* is the most important pathological mechanism of tissue damage in SLE.

Epidemiology of SLE

Ethnic variations and female predominance

- Wide racial differences in the incidence, prevalence, clinical presentation and course of disease
 - Caucasians ~10 - 20/100,000
 - Best disease outcomes amongst different ethnicities
 - Orientals ~50 - 70/100,000
 - Caribbean blacks ~1/250 *women* in Jamaica
 - Worst disease outcomes
- Females : male = 9 : 1

Genetic predisposition

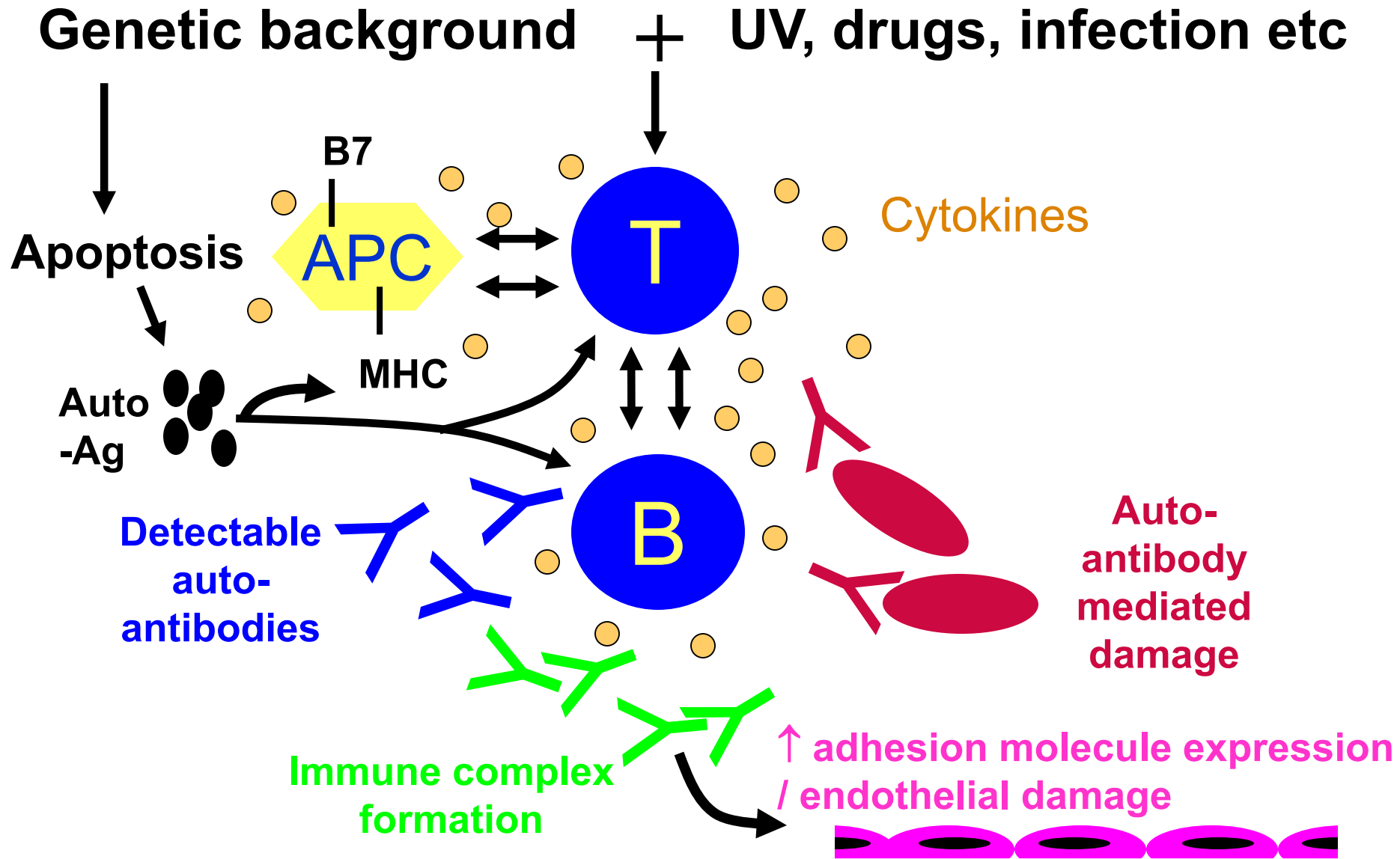
- High concordance rate in identical twins (24% - 70%)
- 10% of relatives have abnormal serology
- Association with HLA B8, DR2 and DR3
- Other HLA III or non-HLA genes
- Genes that influence sex hormone status
- Complement deficiency, e.g. C4 deficiency, occurs in about 5% of patients with SLE

More than just genes

- SLE is common amongst Afro-Americans but rare in Afro-Africans
 - Some infections e.g. malaria may be protective
- Nancy Reagan's thyroiditis and her dog with lupus!!
- Triggering agents
 - Infections e.g. EBV, Parvovirus B19, Rhabdovirus, HIV-1, adenovirus, salmonella may trigger SLE
 - Chemical agents e.g. hydrazine, tartrazine, hair dyes, eosin, heavy metals – mercuric chloride, gold, cadmium
 - Certain foods e.g. L-Canavanine (alfalfa seeds and sprouts)
 - Ultraviolet lights – UVA and UVB

GWAS in SLE

- T cell activation
 - *PTPN22*, *TNFSF4*, HLA class II
- B cell signaling
 - *BANK1*, *BLK*, *PRDMI*
- Immune complex and antigen clearance
 - *ITGAM*, *TREX1*, FcγR genes, complement genes
- TLR-IFN pathways
 - *IRF5*, *STAT4*, *IRAK1*



Failure to remove auto-Abs / ICs = tissue damage

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Clinical domains	Points
<i>Constitutional domain</i>	
Fever	2
<i>Cutaneous domain</i>	
Nonscarring alopecia	2
Oral ulcers	2
Subacute cutaneous or discoid lupus	4
Acute cutaneous lupus	6
<i>Arthritis domain</i>	
Synovitis in at least two joints or tenderness in at least two joints, and at least 30 min of morning stiffness	6
<i>Neurologic domain</i>	
Delirium	2
Psychosis	3
Seizure	5
<i>Serositis domain</i>	
Pleural or pericardial effusion	5
Acute pericarditis	6
<i>Hematologic domain</i>	
Leukopenia	3
Thrombocytopenia	4
Autoimmune hemolysis	4
<i>Renal domain</i>	
Proteinuria >0.5g/24 hr	4
Class II or V lupus nephritis	8
Class III or IV lupus nephritis	10

Immunologic domains	Points
<i>Antiphospholipid antibody domain</i>	
Anticardiolipin IgG >40 GPL or anti-β2GP1 IgG >40 units or lupus anticoagulant	2
<i>Complement proteins domain</i>	
Low C3 or low C4	3
Low C3 and low C4	4
<i>Highly specific antibodies domain</i>	
Anti-dsDNA antibody	6
Anti-Smith antibody	6

EULAR / ACR 2019

- ANA $\geq 1/80$
- SLE = ≥ 10 points
- At least one clinical domain
- Exclude other more likely disease
- No double count within each domain; domain with highest point only
- Criteria can be serial or in parallel

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SLE - Constitutional symptoms

- Fever
- Malaise
- Poor appetite
- Weight loss

SLE - mucocutaneous features

- Photosensitivity
- Malar (butterfly) rash
 - May or may not be associated with sun exposure
 - Nasolabial folds often spared
- Discoid lupus erythematosus
- Subacute cutaneous lupus
- Cutaneous vasculitis
 - nail edge / fold, splinter haemorrhages, palm, elbow, lower limb
- Livedo reticularis
- Hair loss
 - May be scarring or non-scarring

Malar rash (Butterfly rash)





Subacute cutaneous LE

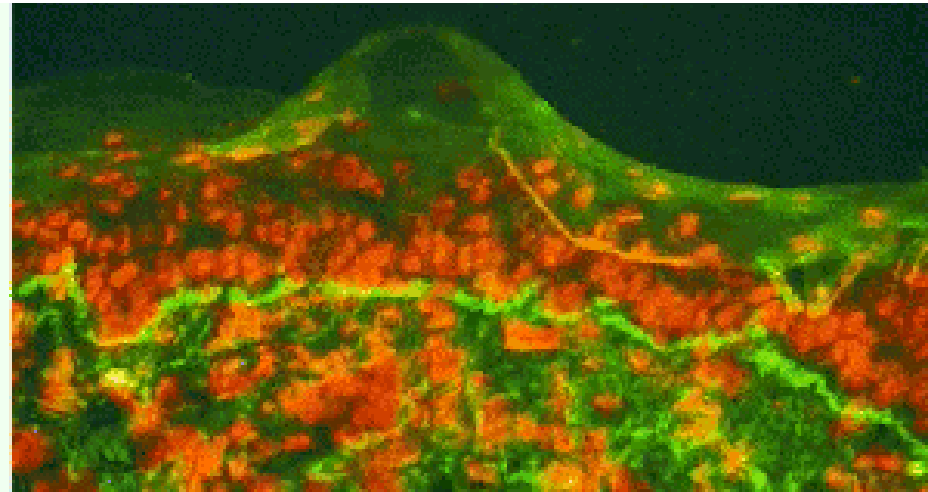
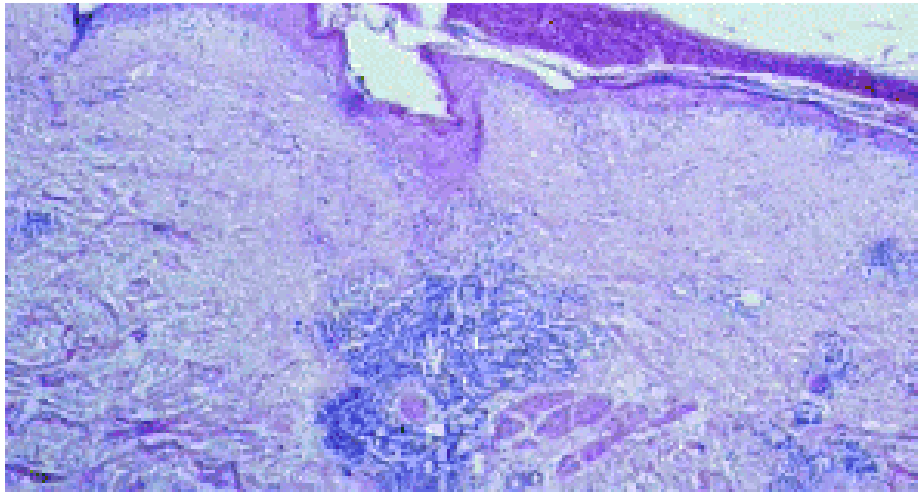


Chronic discoid LE

Non-scarring alopecia in SLE



Cutaneous lupus – lupus band test



Lupus band: deposition of immune complexes

Vasculitic lesions



SLE – other mucocutaneous features

- Purpuric skin rash
 - thrombocytopaenia
 - leucocytoclastic vasculitis
 - Differentiating features from thrombocytopaenic purpura: Raised, papula and sometimes tender
- Erythema nodosum
- Oral ulcers
- Raynaud's phenomenon

Mouth ulcers in a patient with SLE



SLE - Musculoskeletal features

- Non-deforming polyarthritis/polyarthralgia
 - RA distribution but no radiological erosions
- Avascular necrosis
- Deforming arthropathy - Jaccoud's arthritis
- Erosive arthritis - rare
- Myopathy - weakness, myalgia & myositis
 - May be drug induced or SLE complications
- Myasthenia gravis

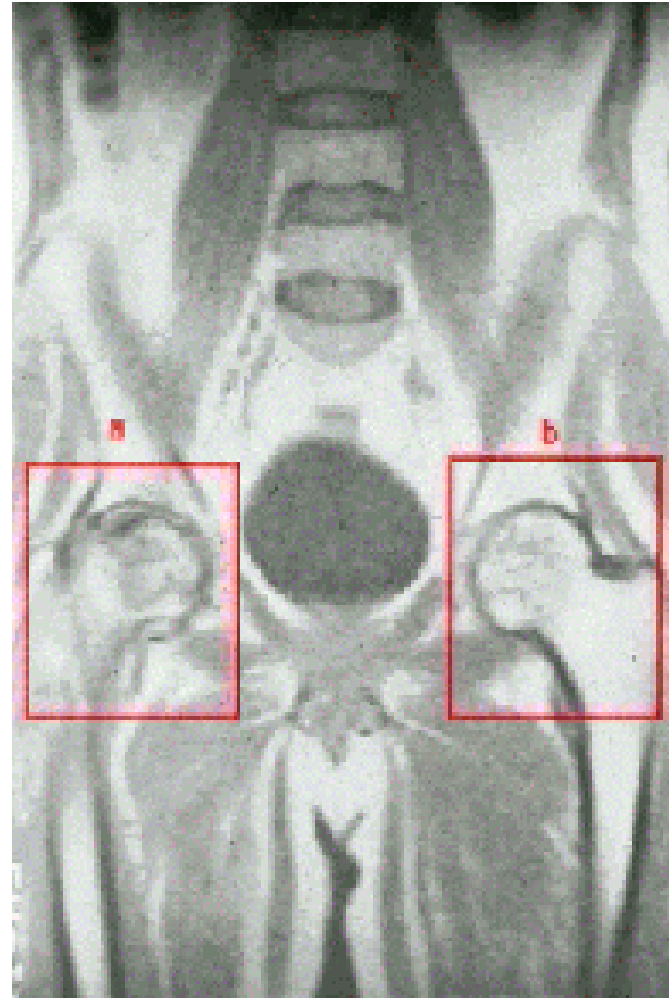


Non-deforming, non-erosive arthritis



Jaccoud's arthritis:
Deforming, non-erosive arthritis

Avascular necrosis



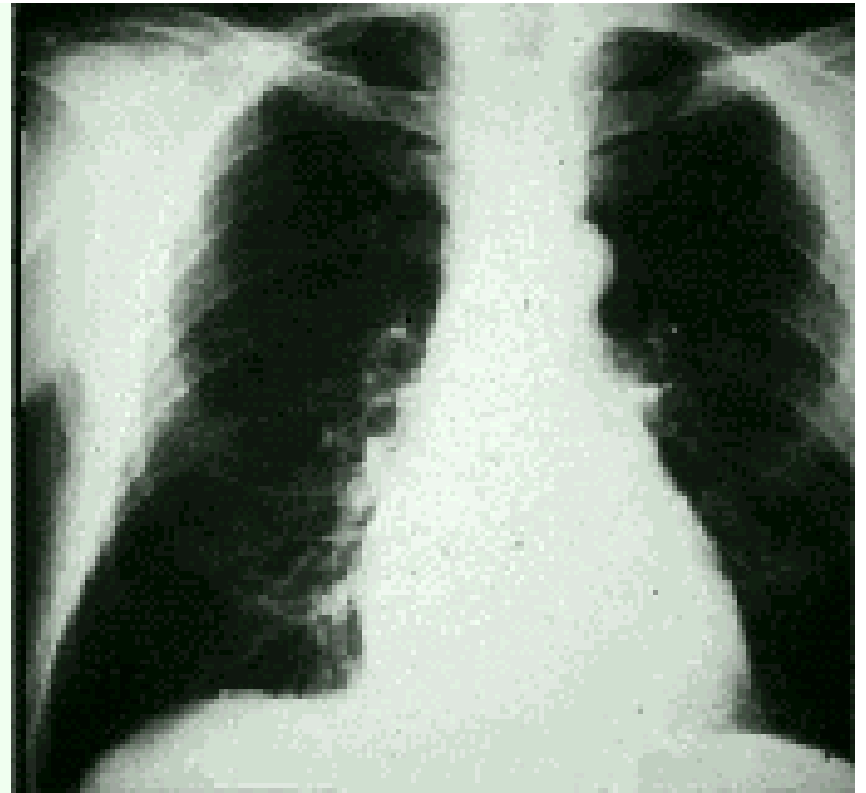
SLE - Pulmonary features

- Pleurisy
- Infections
- Recurrent atelectasis
 - shrunken lungs
- Diffuse lung infiltration and fibrosis
- Pulmonary hypertension
- Pulmonary infarct

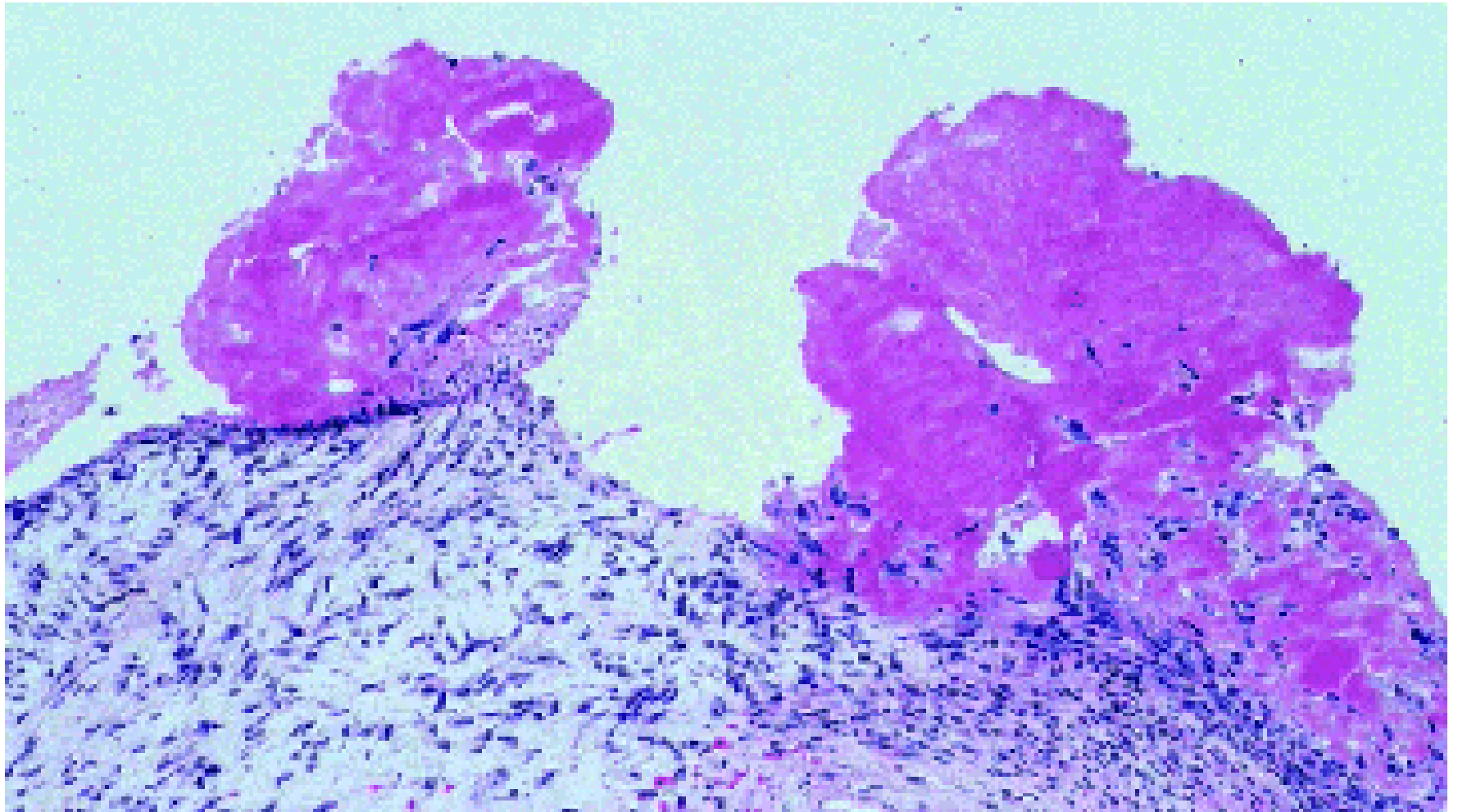
SLE - Cardiac features

- Pericarditis
- Cardiomyopathy
- Pulmonary hypertension
- Libman Sach endocarditis

Development of pericardial effusion in a patient with SLE

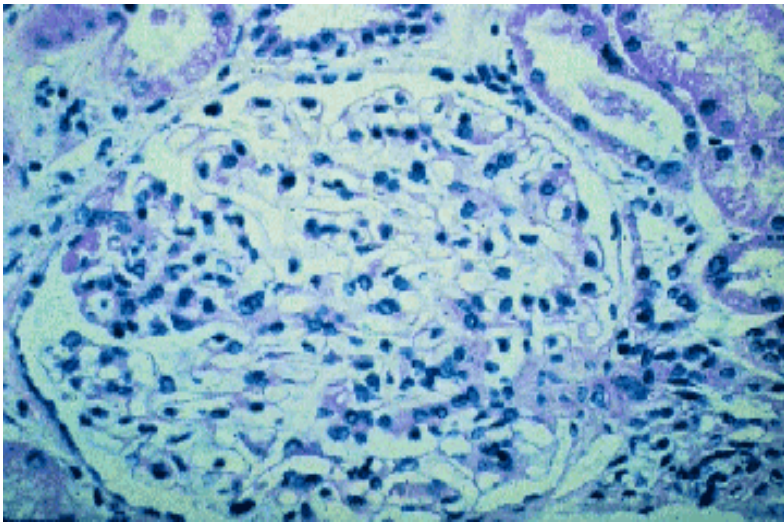


Libman Sach aseptic endocarditis

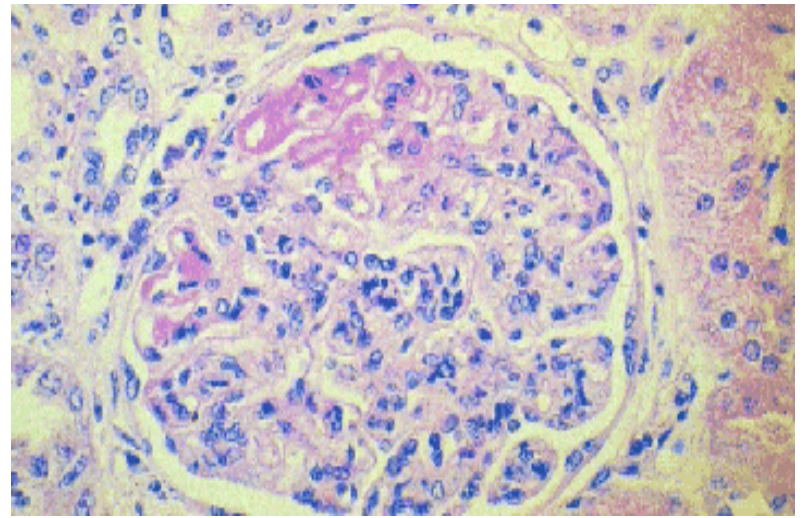


SLE - Glomerulonephritis

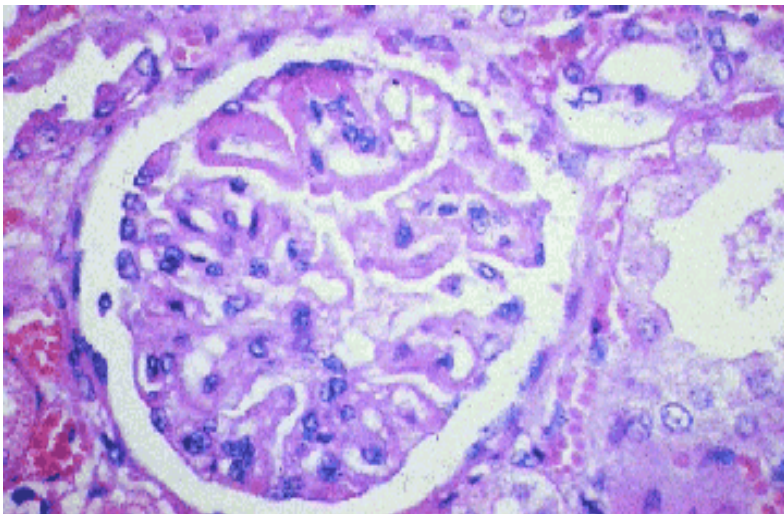
- Presentation
 - **Proteinuria**
 - Urine sediments
 - Urine RBC and casts
 - Hypertension
 - Acute renal failure
 - Chronic renal failure
- WHO class I - VI
 - I = Normal
 - II = Mesangial
 - III = Focal proliferative
 - IV = Diffuse proliferative
 - V = Membranous
 - VI = Sclerotic



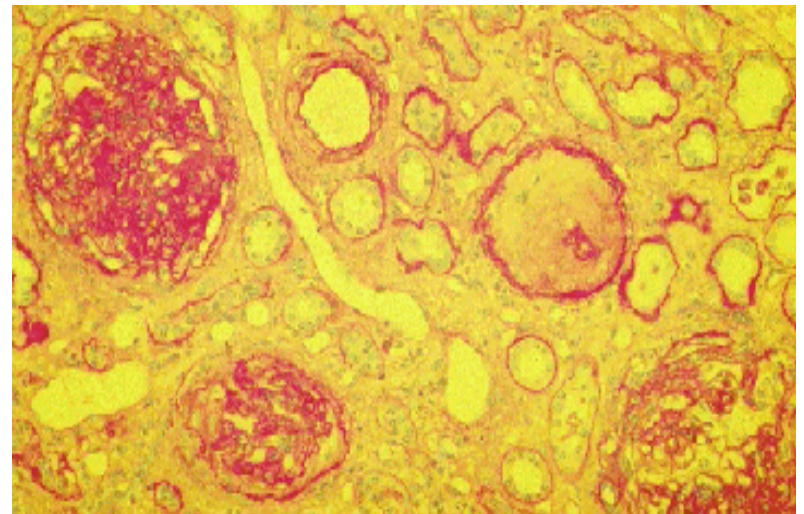
III – focal proliferative



IV – diffuse proliferative

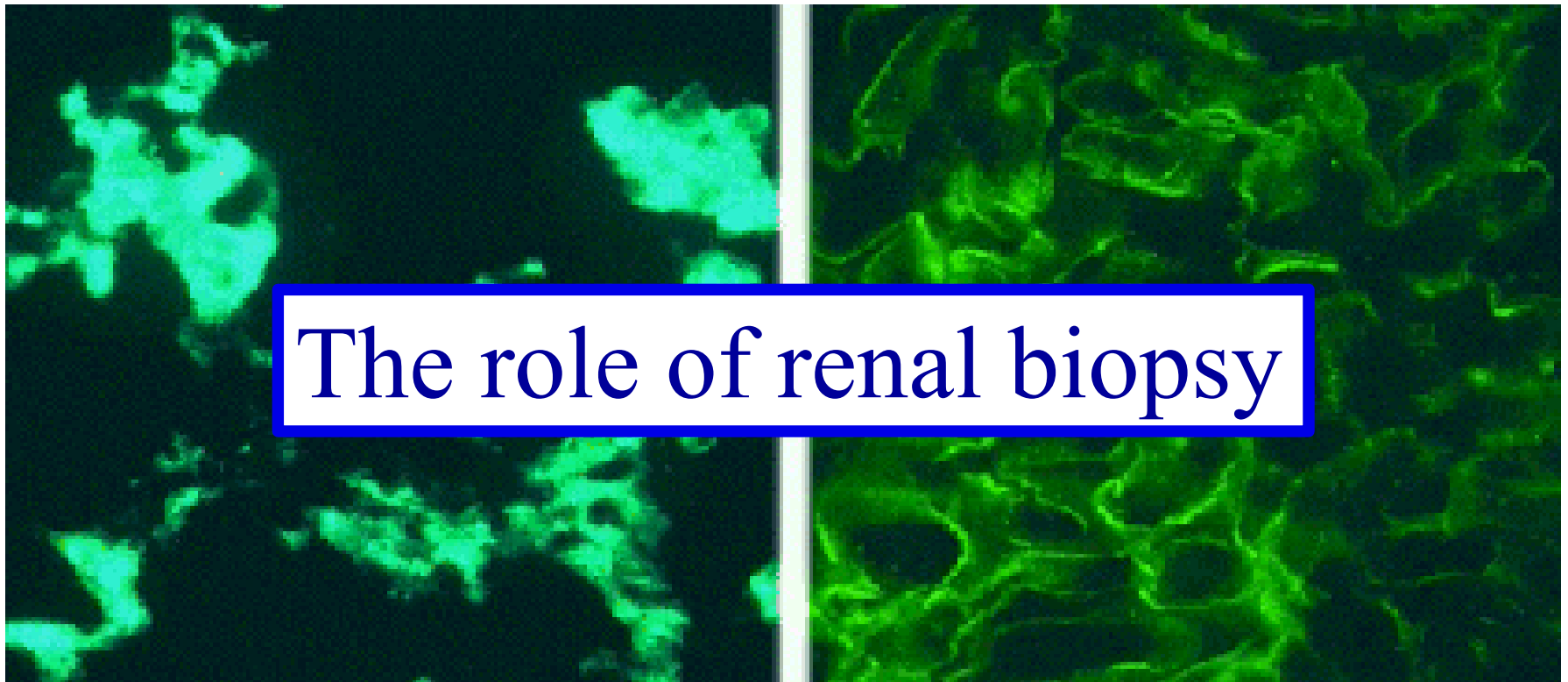


V – Membranous



VI - sclerosis

Immuno-staining showing Ig deposition



SLE - Neurological features

- Depression/psychosis
 - Not always related to disease activity
 - Considerable variability
 - Usually reversible
 - Somatic treatment may help
- Migranous headache
 - Peri-menstrual
 - May be associated with disease activity
 - Poor response to treatment

Patho-aetiology:

Unknown

SLE - Neurological features

- Cerebral ischaemia
 - TIAs or stroke
- Retinopathy
- Cranial or peripheral neuropathy
- Myelitis
- Chorea
- Cerebellar ataxia
- Meningitis and cerebral abscesses

Patho-aetiology:

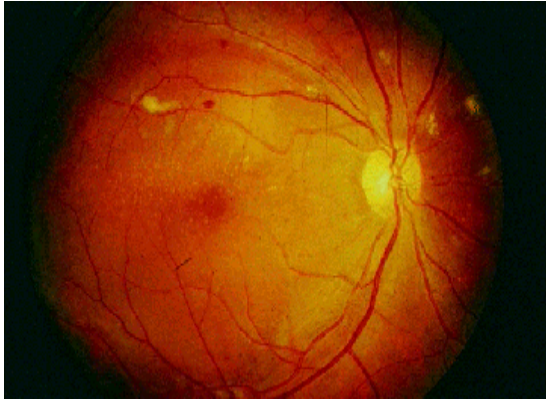
Vasculitis

Thrombosis

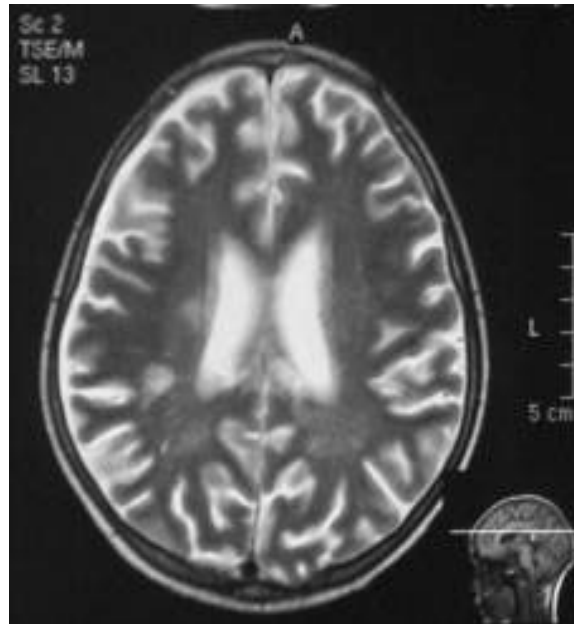
Auto-Ab mediated

Infection related

Unknown



Cytoid bodies



Microinfarcts

SLE - Haematological features

- Lymphadenopathy
 - ~25% of all patients during their course of illness
- Leucopaenia
 - Anti-leucocyte antibodies
- Anaemia
 - Haemolytic anaemia
 - Normochromic normocytic
- Thrombocytopaenia
 - Anti-platelet antibody
 - Anti-phospholipid syndrome

SLE - Haematological features

- Thrombosis
 - Anti-phospholipid antibody syndrome
 - Vasculitis
 - Use of corticosteroids
- Splenomegaly

SLE - Susceptibility to infection

- Intrinsic factors
 - Low complements
 - Anti-WBC antibodies
 - Impaired cell mediated immunity
 - Defective phagocytosis
 - Poor antibody response to certain antigens
 - Hyposplenism
- Extrinsic factors
 - Steroids
 - Other immunosuppressive drugs
 - Nephrotic syndrome
 - Uraemia

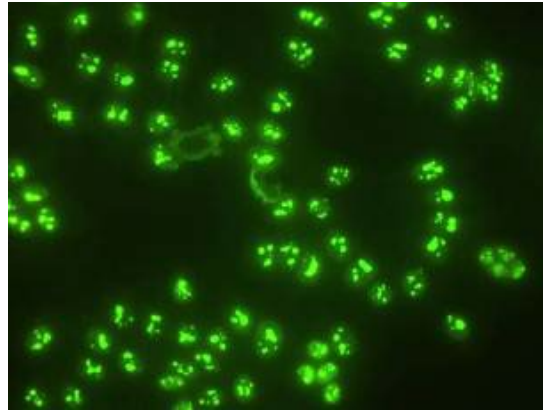
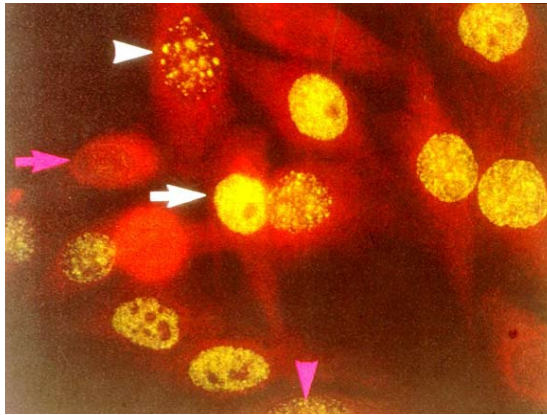
Infections: the most important cause of death in SLE

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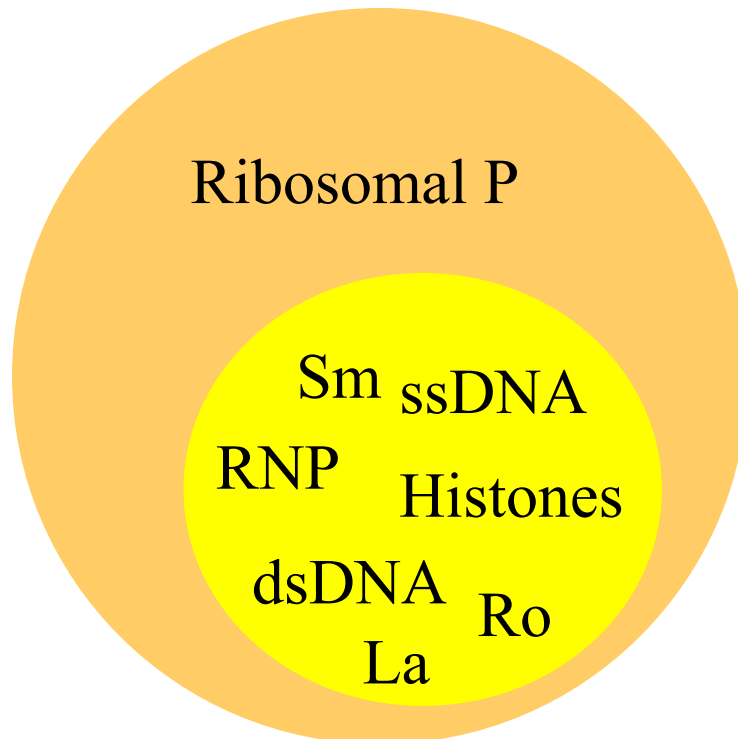
Screening tests for suspected SLE

- ANA
 - Usually suffices
 - Anti-dsDNA, anti-ENA and APA only if ANA is +ve or SLE is strongly suspected
- Rheumatoid factor
 - To exclude possible RA / Sjogren's
- Complement levels
 - Depressed during active disease
- Ig pattern if strong degree of suspicion
- Complete and differential blood counts
- Renal function tests including urine examination
- Serum levels of albumin and globulin



ANA

- Anti-ssDNA
- Anti-histone
- Anti-dsDNA
- Anti-ENA
 - Anti-Sm
 - Anti-RNP
 - Anti-Ro
 - Anti-La
 - Anti-P



Anti-nuclear antibodies

- They are a group of auto-antibodies
- **Highly sensitive** but **non-specific** and not diagnostic
- Staining pattern has no prognostic values
 - Homogenous, speckled, nucleolar etc
- Serum titre does not correlate with disease severity and activity
- May become negative if there is severe proteinuria

SLE – common auto-antibodies

- Anti-dsDNA antibodies (60% – 70%)
- Anti-ENA antibodies
 - Anti-Ro (40 – 50%) +/- anti-La (30% – 40%)
 - Anti-Sm (10% - 20%)
 - Anti-RNP (30% - 40%)
- Antiphospholipid antibodies
 - Anti-cardiolipin antibodies (IgG and IgM)
 - Anti- β 2 glycoprotein I antibodies
 - Lupus anticoagulant (coagulation assay)

Anti-dsDNA

- Occurs in ~60% of patients with SLE
- Highly **specific** for SLE
- Is one of the classification criteria for SLE
- Titre correlates with overall **disease activity**
- May be associated with lupus nephritis

Anti-ENA in SLE

- A group of different autoantibodies
- Most, except anti-Sm, are not diagnostic of SLE
- Probably no correlation with disease activity
- Found in many connective tissue disorders
- May **predict clinical manifestations**
- May have **prognostic** value

Anti-phospholipid syndrome

- Positive anti-cardiolipin antibodies and / or lupus anticoagulant
- Arterial / venous thrombosis
- Superficial thrombophlebitis and livedo reticularis
- Transverse myelitis
- Recurrent abortions
- Mild/moderate thrombocytopenia

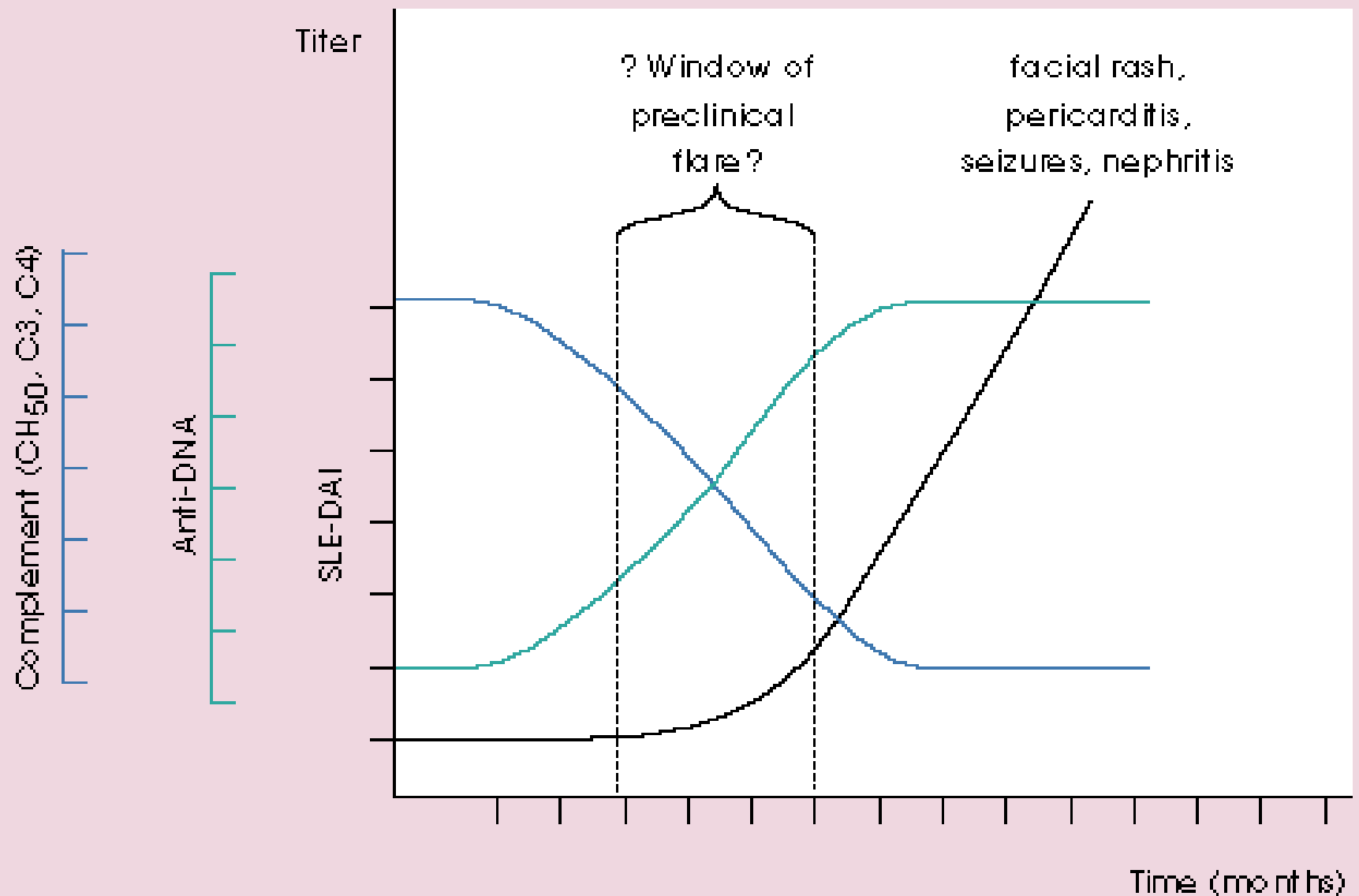
1° vs 2° Anti-phospholipid syndrome

- 1° APLS
 - Anti-PL antibodies
 - Recurrent thrombosis
 - Superficial thrombophlebitis
 - Transverse myelitis
 - Recurrent abortions
 - Mild / moderate thrombocytopenia
 - NO UNDERLYING CONDITIONS
- 2° APLS
 - Features of APLS
 - AN UNDERLYING CONDITION
 - SLE
 - Other connective tissue disorders
 - Patients with other conditions may have anti-PL but not thrombosis etc. They do not have APLS

Monitoring SLE activity

- Thorough clinical assessment including BP
- ANA and anti-ENA have no role
- Anti-dsDNA level positively correlates with activity
- C3/C4 levels negatively correlate with activity
- Urine examination including protein, cells and casts
- Complete blood count
- Blood biochemistry

IMMUNE STUDIES AND SLE DISEASE ACTIVITY



SLE and CRP

- In most cases, active SLE disease does not evoke a CRP response
- High CRP levels in patients with SLE should lead to suspicion of infection
- Patients with active arthritis, serositis or vasculitis may have moderately elevated CRP

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SLE - General management

- Counselling - patients, spouse and relatives
 - Poor compliance is a major risk factor of poor prognosis
- Regular monitoring
- Avoid excessive sun-exposure
- Avoid potential antigenic stimuli
- ? Avoid oral contraceptives
- Infection control
- Pregnancy issues
- Supportive treatment e.g. dialysis

Four overarching principles

1. SLE is a multisystem disease. Diagnosis is based on clinical grounds + serologic abnormalities
2. SLE care is multidisciplinary, based on a shared patient-physician decision
3. Organ-/life-threatening complications require initial high-intensity and subsequent maintenance immunosuppressive therapy
4. Treatment goals include long term patient survival, prevention of organ damage and optimisation of health-related quality of life

Drug treatment - 4 recommendations

1. Goals of treatment
2. Treatment of SLE (General)
3. Specific manifestations
4. Comorbidities

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Drug treatment - 4 recommendations

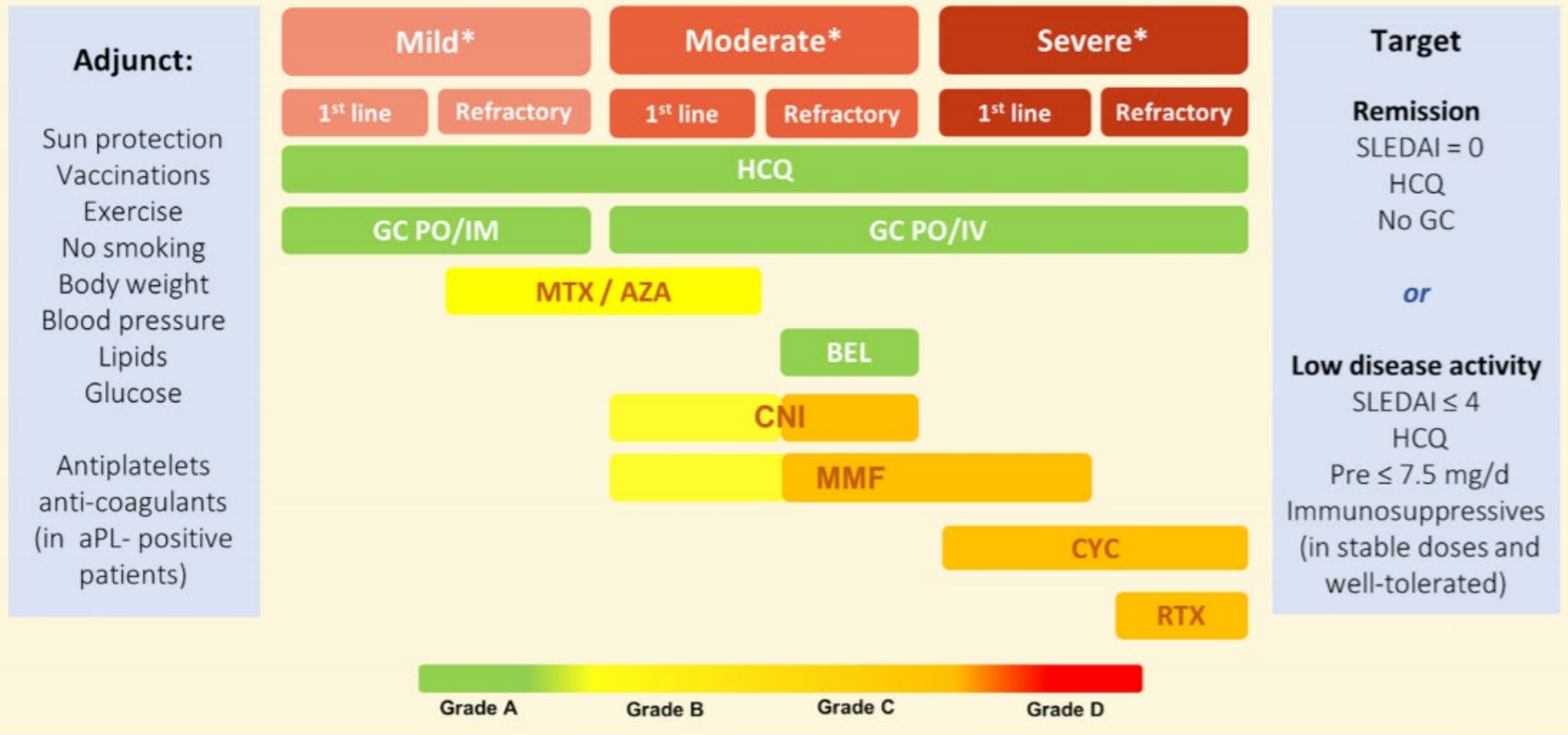
1. Goals of treatment

- Aim at remission or **low disease activity** and prevention of flares in all organs
- **Lowest possible dose of glucocorticoids**
- Flares are to be treated according to the severity of organ(s) by adjusting ongoing therapies

Drug treatment - 4 recommendations

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Treatment of non-renal Systemic Lupus Erythematosus



Mild: constitutional symptoms/ mild arthritis/ rash ≤9% BSA/PLTs 50-100 x 10³/mm³; SLEDAI≤6; BILAG C or ≤1 BILAG B manifestation

Moderate: RA-like arthritis/ rash 9-18% BSA/cutaneous vasculitis ≤18% BSA; PLTs 20-50x10³/mm³/serositis; SLEDAI 7-12; ≥2 BILAG B manifestations

Severe: major organ threatening disease (nephritis, cerebritis, myelitis, pneumonitis, mesenteric vasculitis; thrombocytopenia with platelets <20x10³/mm³; TTP-like disease or acute hemophagocytic syndrome; SLEDAI>12; ≥1 BILAG A manifestations

Drug treatment - 4 recommendations

1. Goals of treatment

2. Treatment of SLE (General)

- **Hydroxychloroquine** is recommended for all
- Use of glucocorticoids
- Immunosuppressive therapies
- **Biologics** – belimumab and rituximab; anti-IFN-alpha

Hydroxychloroquine

- Indicated for almost all patients
- Useful for arthritis, cutaneous manifestations and constitutional symptoms
- May reduce systemic complications
- May improve pregnancy outcomes
- May reduce secondary infectious disease complications

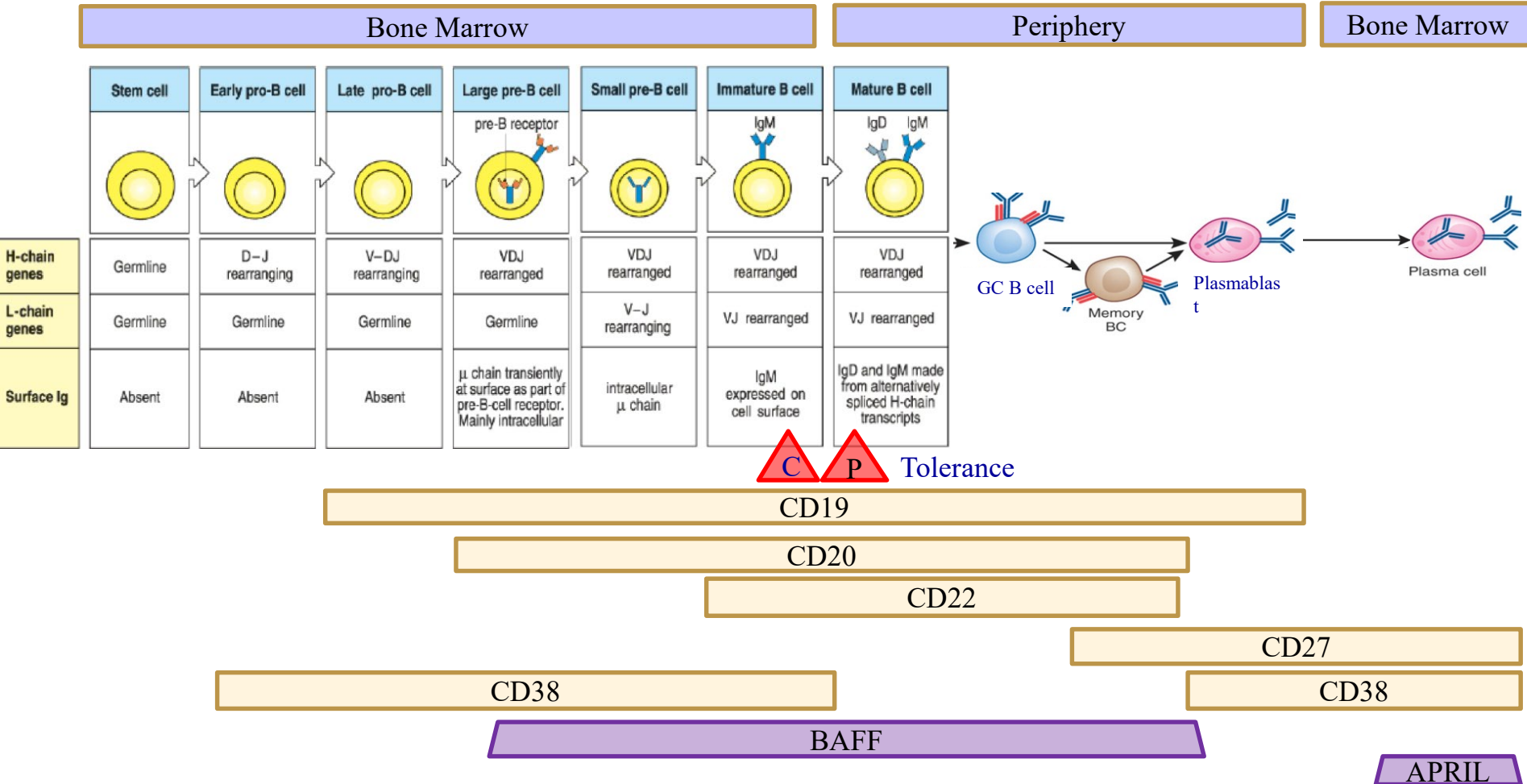
Hydroxychloroquine

- Drug blood levels to assess compliance
- >10% retinal toxicities after 20 years of use
 - Major risk factors
 - Duration of treatment (OR 4.71 for every 5 years)
 - Dose (OR 3.34 for every 100 mg daily dose
 - Risk is very low for doses <5 mg/Kg/day
 - Chronic kidney disease (Adjusted OR 8.56)
 - Pre-existing retinal or macular disease

Immunosuppressive agents

- Azathioprine
 - Hepatotoxicity
 - Cyclophosphamide
 - Infertility
 - Haemorrhagic cystitis
 - Cyclosporin A
 - Renal toxicity
 - Methotrexate
 - Hepatotoxicity
 - Leflunomide
 - Hepatotoxicity
 - Mycophenolate mofetil
 - Tacrolimus
- All can cause bone marrow suppression
 - All can cause increased susceptibility to infection
 - All potentially carcinogenic
 - All potentially teratogenic

Targeting B cells – belimumab and rituximab



SLE and type I interferon

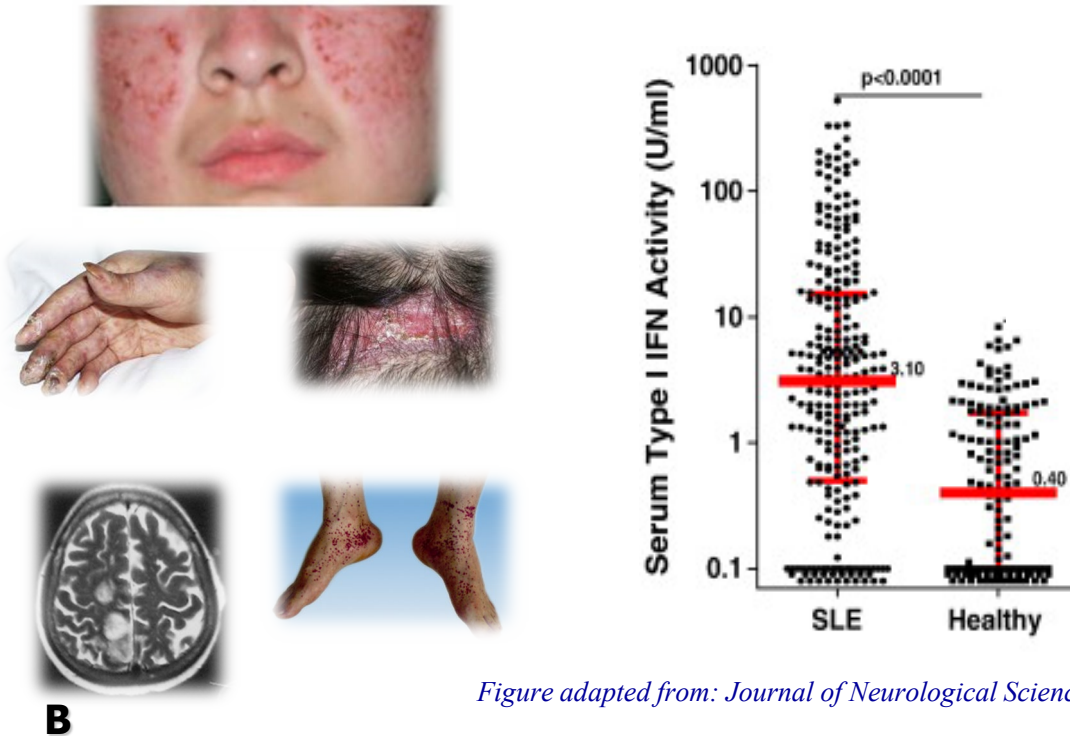
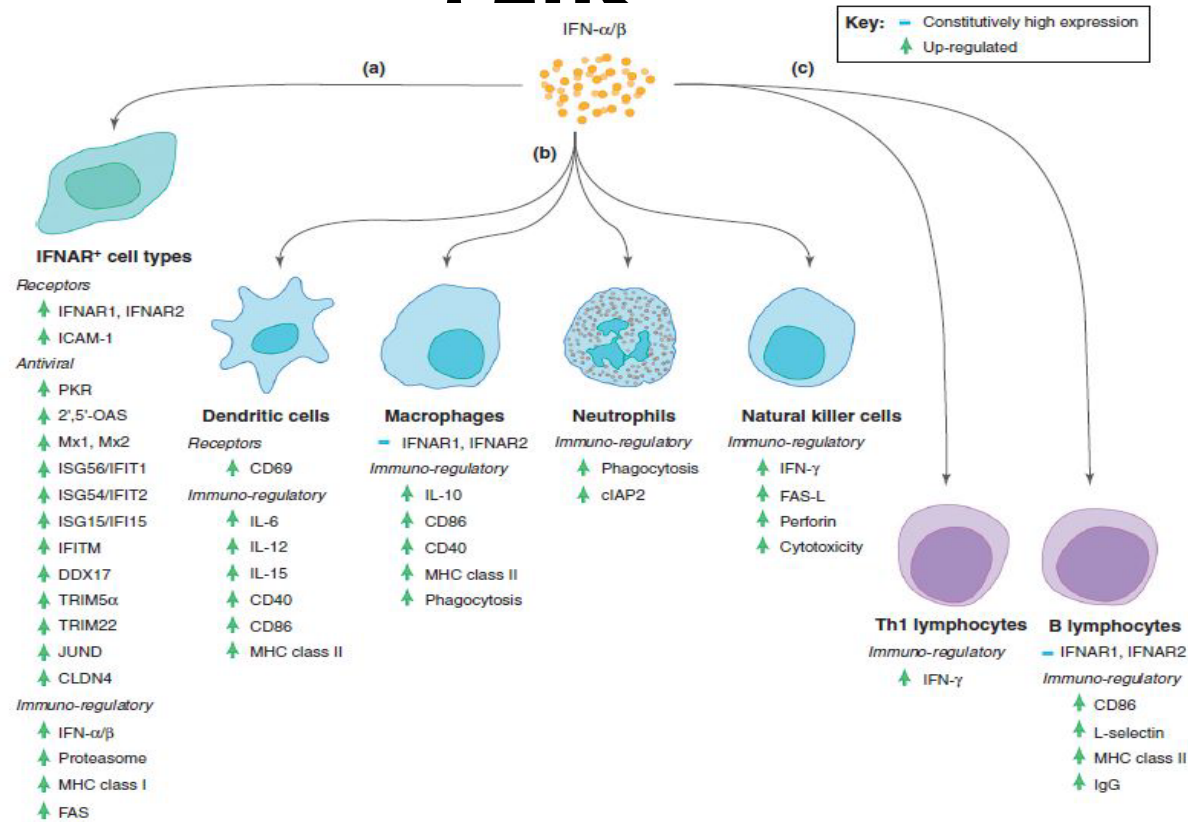


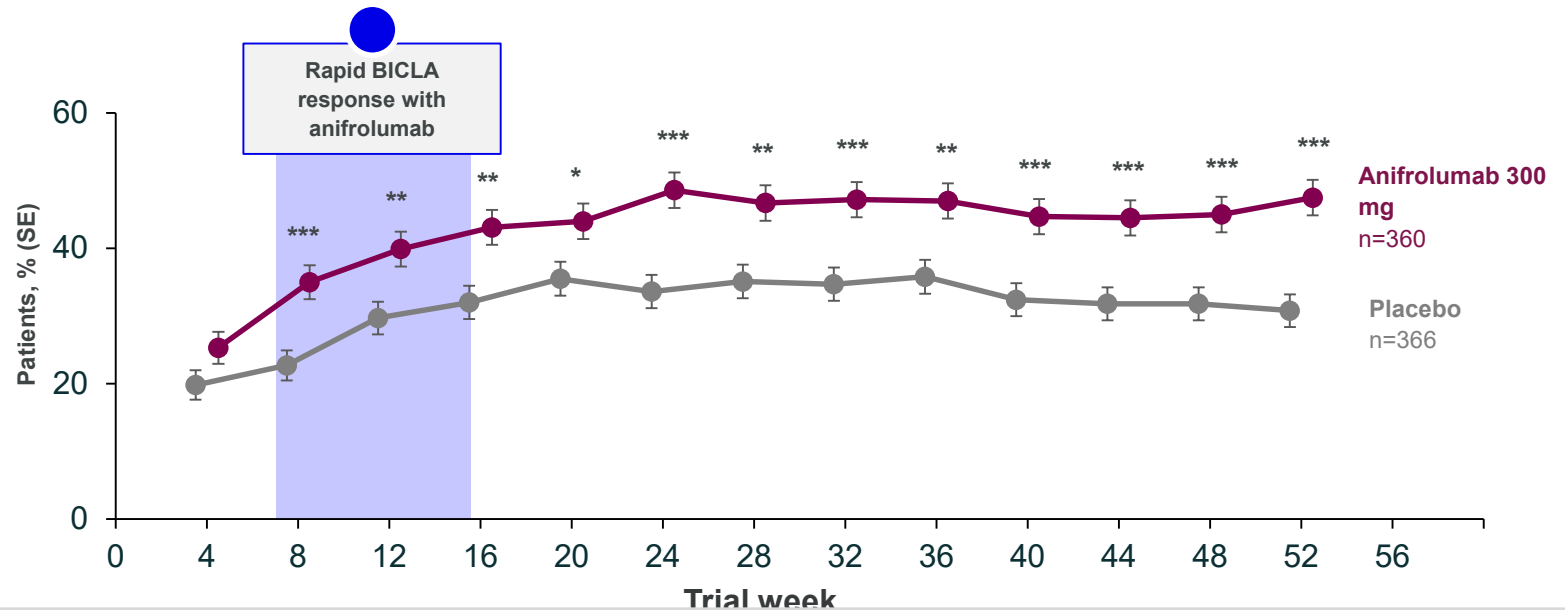
Figure adapted from: *Journal of Neurological Sciences*, Feng et al, 2012

IFN α network and immune cells





TULIP 1&2 BICLA responders



More patients showed a BICLA response as early as W8 and W12 in the anifrolumab group vs placebo, a significant difference that persisted over the duration of the trial²

BICLA, British Isles Lupus Assessment Group-based Composite Lupus Assessment; SE, standard error; W, Week.

Restricted medication rules were according to the TULIP-2 protocol. The response rates were adjusted for the factors which randomization was stratified using the stratified Cochran-Mantel-Haenszel method. Nominal *P*-values are presented, **P*<0.05; ***P*<0.01; ****P*<0.001.

1. AstraZeneca Data on File. 2. Morand EF, et al. Presentation OP0003. Presented at: EULAR Virtual Congress; June 2020.

Drug treatment - 4 recommendations

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Drug treatment - 4 recommendations

1. Goals of treatment
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3. Specific manifestations
 - Skin
 - Neuropsychiatric disease
 - Haematological disease
 - Renal disease

Drug treatment - 4 recommendations

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Drug treatment - 4 recommendations

1. Goals of treatment
2. Treatment of SLE (General)
3. Specific manifestations
4. Comorbidities
 - Antiphospholipid syndrome
 - Infectious diseases
 - Cardiovascular disease

Lecture outline

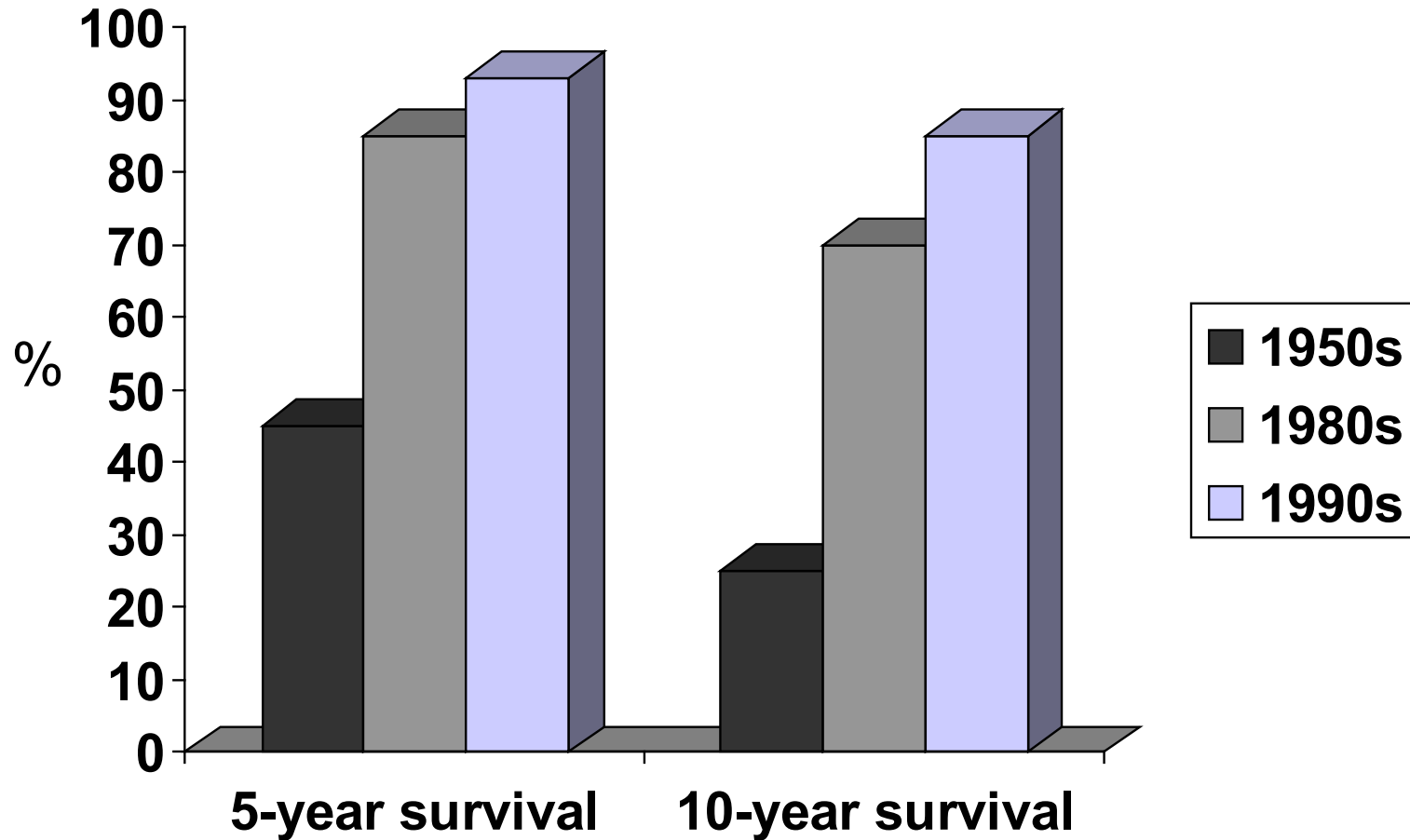
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Living in the shadow of lupus



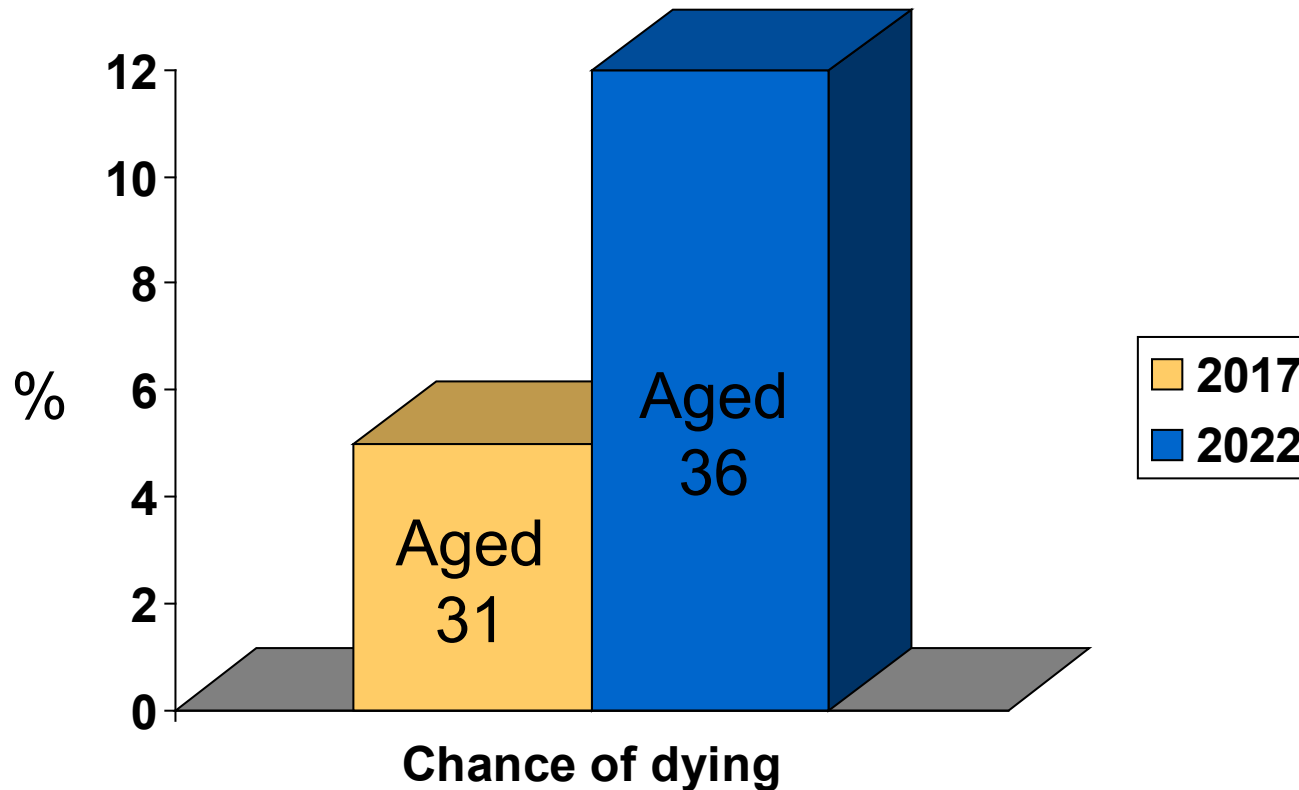
Once upon a time there were 3 little pigs.
So I ate them. End of story.

Lupus long term survival



What do these figures mean?

- The chance of Helen dying 5 and 10 years on!



What do patients with SLE die of?

- Complications of SLE in the first few years
 - Risk factors
 - Late disease onset
 - Male SLE
 - Lower socioeconomic class
- Infection is the most important cause of death in the early / medium stage of disease
- Cardiovascular thrombosis is the major the cause of death in the late stage





TELL US WHAT

JC Whole Class Session

YOU THINK

THANK YOU!



**Look for today's teacher
and the others you may have missed.**

(HKU Portal login required)